

ApeiroUI – a Visual Interface for Cloud Infrastructure

Bridging Fragmentation in Cloud DevOps Through Designing a Unified Visual Infrastructure Interface

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Abstract

Cloud and bare-metal infrastructure management is increasingly hampered by tool fragmentation and a lack of unified user experience. System administrators and DevOps engineers deal with an abundance of disconnected dashboards, CLIs, and scripts, making it challenging to maintain a coherent mental model of the system. This paper presents the design of ApeiroUI, a conceptual user interface for cloud infrastructure that aims to integrate management of cloud services, on-premises hardware, and user access into a single, human-centered interface. Developed through a collaboration between SAP and the University of Applied Sciences Potsdam, ApeiroUI was shaped using co-design methods. In two co-design workshops with developers and designers, we analyzed workflows and use cases that directly influenced the user interface design. The resulting prototype provides a unified visual language and introduces a novel context-based AI assistant. We describe the co-design methodology, key design decisions, and the lessons learned from this industry-academic project. Our case study demonstrates how a design-led approach can address DevOps tool fragmentation, improve infrastructure usability, and illustrate the value of industry-academic collaborations.

CCS Concepts

• **Human-centered computing** → **User interface design; Participatory design**; • **Computer systems organization** → *Cloud computing*.

Keywords

Co-design, user interface design, cloud computing

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1 Project Setting: A Cross-Sector Collaboration

Modern cloud environments suffer from extensive tool fragmentation. Teams rely on numerous dashboards, Command Line Interfaces (CLI), and monitoring tools. Development and Operations (DevOps) engineers frequently switch contexts, manually correlate information, and maintain mental models across disparate systems—an error-prone and cognitively demanding process. Industry trends, including findings from the 2024 DORA report [1], highlight a growing need for unified internal platform tooling. These developments underscore a shared need: coherent, integrated user experiences for cloud operations.

The ApeiroUI project addresses these challenges. It resulted from a collaboration between SAP and the University of Applied Sciences Potsdam (FHP) within SAP's contribution to the broader IPCEI-CIS¹ initiative on next-generation cloud infrastructure called ApeiroRA.² SAP contributed industry expertise in large-scale Kubernetes operations, hardware provisioning, and DevOps workflows, while FHP brought experience in design research, interaction design, and participatory methods. This cross-sector setup ensured that ApeiroUI addresses real operational challenges while grounding the design in human-centered principles.

2 Co-Designing ApeiroUI: Methods and Workshop Insights

To rethink the user experience of Cloud DevOps, this project relies substantially on Participatory Design (PD) as a methodology. PD has its roots in the Scandinavian design movement of the 1970s–80s, emphasizing co-creation with users and democratization of design decisions [3, 4]. More recently, PD has been applied in domains such as software engineering tools and data visualization. Wacnik, Daly, and Verma's systematic review [5] shows that participatory design is increasingly used in complex and technical contexts, not just consumer product design. They highlight that successful PD in such contexts often uses tangible artifacts and scenarios to bridge communication between domain experts and designers. Additionally, prior work by Dörk, Müller et al. [2] demonstrates how co-designing visualizations with domain experts can surface tacit knowledge and produce interfaces that better match users' mental models.

We conducted two co-design workshops with mixed groups of engineers and designers (about eight participants per workshop). Participation in the workshops was voluntary. For the hands-on

¹Important Project of Common European Interest on Next-Generation Cloud Infrastructure and Services (IPCEI-CIS).

²<https://apeiora.eu/>

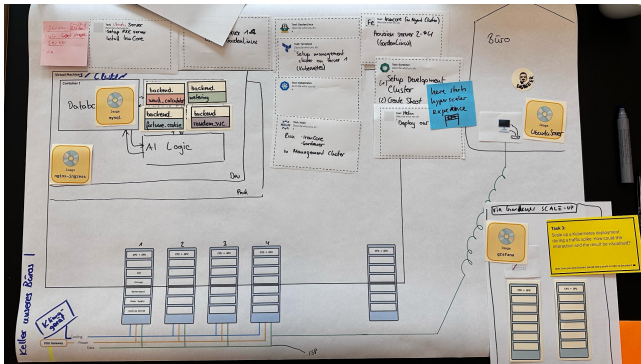


Figure 1: Result from the first co-design workshop. The board describes the initial setup process for cloud hardware. Note the use of stencils and tokens.

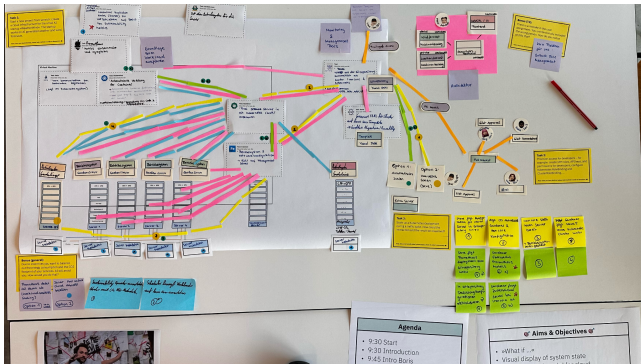


Figure 2: Result from the second co-design workshop. The board visualizes software dependencies within a Kubernetes cluster. Note the use of stencils, tokens, post-its, and connecting line elements.

activities, each engineer was paired with a designer to combine operational expertise with interface thinking. The focus of the workshop assignments was on regular DevOps workflows and the challenges of tool fragmentation.

Workshop 1 centered on hardware provisioning and datacenter workflows: bringing bare-metal resources online, monitoring them, associating them with projects, and linking them to cloud layers. Workshop 2 focused on cloud orchestration and application operations: deploying, scaling, observing, and investigating services.

To support collaboration, we prepared tangible materials: schematic tokens for typical entities (servers, containers, databases, users, UI elements) and task cards describing realistic scenarios. DevOps engineers described how they currently perform the tasks, then the team sketched how a visualization or a user interface model could support them. The physical format helped participants make dependencies explicit and compare alternative representations (Figures 1 and 2).

Both sessions revealed a common pattern: operational understanding is generated by constantly combining diverse workflows.

Once the use cases were mapped, participants proposed interaction design concepts that prioritized integration, continuity, relationships, and fast movement between levels—from service to node, from incident to owner, or from hardware asset to affected workload.

From these workshops, we distilled a few key design principles:

- **Integrate, don't replace:** The interface should pull together data and controls from existing tools (APIs, CLIs) rather than force users to abandon them.
- **Make relationships visible:** The interface should reveal dependencies between services, workloads, hardware, costs, and responsibilities.
- **Contextual assistance, not clutter:** Intelligent support is valuable when it is tied to the current object or task and stays unobtrusive.
- **Consistency across views:** Multiple perspectives should feel like one system through shared navigation, terminology, and visual conventions.

3 Prototyping ApeiroUI: Visual Language and Key Components

Based on these key design principles, a Figma-based prototype was created to demonstrate the core features.³ While not functionally integrated, the prototype includes realistic interactions and visualizations that reflect the design principles. The prototype is structured into three main views and is integrated via a consistent user interface that provides the users with direct access to current projects.

COS (Cloud Operating System) View: The COS view (Figure 3) presents cloud-native components such as applications, services, and health indicators. It consolidates tools commonly used for deployments, logs, service dependencies, and scaling workflows. A dedicated tool-stack panel allows teams to add or remove open-source cloud tools, supporting a modular, extensible ecosystem.

BOS (Bare-Metal Operating System) View: The BOS view (Figure 4) focuses on physical infrastructure—data centers, racks, servers, networks, and resource usage. The BOS view includes visual rack maps, asset metadata, and quota overviews, along with access to Asset Administration Shell (AAS) data for detailed hardware specifications.

People View (Access & Collaboration): The People view provides an overview of users, teams, and roles across the infrastructure. By clarifying responsibilities and permissions, the view supports collaboration and security oversight. Engineers can identify system owners, troubleshoot more efficiently, and understand how access is structured.

Context-aware AI Assistant: A key innovation is the context-aware AI assistant, accessible via a floating button. Users can drag the assistant onto interface elements—such as a server, a failing service, or a user profile—to ask questions or initiate operations. This blends conversational AI with spatial interface cues, grounding AI interactions in the user's current workflow.

The prototype includes roughly a dozen screens illustrating typical interactions. Early evaluations—conducted through demos and a

³Video prototype: https://sapvideo.cfapps.eu10-004.hana.ondemand.com/?entry_id=1_0j0esavr

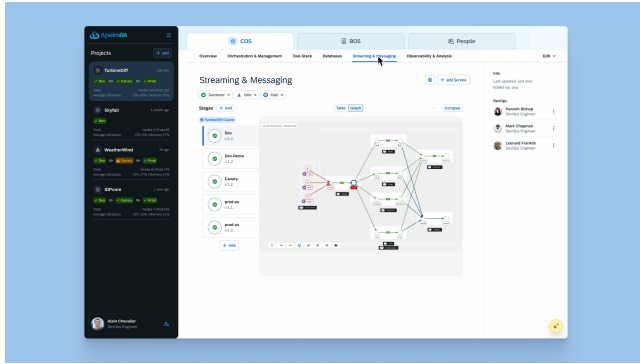


Figure 3: Screen from ApeiroUI showing the Cloud Operating System (COS) view.

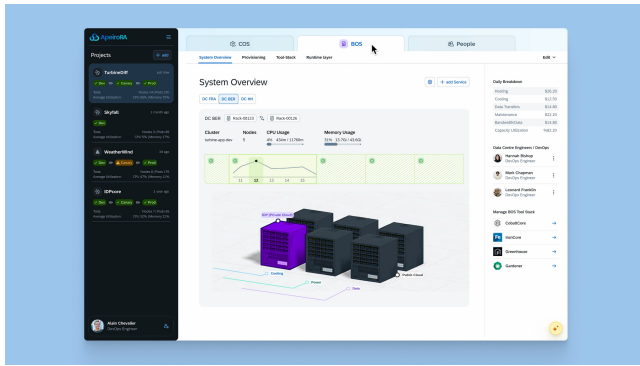


Figure 4: Screen from ApeiroUI showing the Bare-Metal Operating System (BOS) view.

small questionnaire—indicate strong interest in unified tooling and positive responses to the assistant’s contextual interaction model. The evaluations followed the ethics regulations of the University of Applied Sciences Potsdam. Participants emphasized that the prototype improved their ability to maintain an overview of complex systems. To quote one participant: “it would be great to have all the tooling in one place!” Based on this feedback, we intend to develop a more functional prototype and gather more insights.

4 Conclusion: Designing for Human-Centered DevOps

ApeiroUI demonstrates how co-design workshops and an attentive design process can stimulate innovation in a highly technical work environment like cloud development and operations. The fragmented experience of operating hybrid, cloud-native systems with many specialized tools provided numerous opportunities for UI design improvements and innovations. Early feedback indicates both a need and an openness of DevOps engineers for greater integration of cloud management tools through a unified visual user interface without discarding existing toolchains. The inclusion of a context-based AI assistant further shows how emerging technologies improve usability—not by replacing human operators, but by supporting them with context-aware information and actions.

Acknowledgments

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